



DEPLOYMENT SOLUTIONS

SITE ASSESSMENT FIELD GUIDE

Drone dock deployment — where a dock can go, what the site must provide, and what to watch for. Complete one pass per candidate dock location.



READ THIS FIRST

HOW TO USE THIS GUIDE

This guide is for every DXD field rep and sales rep who walks a customer site ahead of a dock deployment. It follows the assessment in order: desk work before you leave, siting and structural checks on the ground, the power and network spec the site must provide, and the red flags that stop a deployment. Work it top to bottom — one full pass per candidate dock location, even when the customer has already picked a spot.

Two things to hold onto. First, you are assessing a location, not selling one — a candidate spot that fails is a finding, not a failure; move the dock and assess again. Second, everything you observe becomes evidence — the deployment is engineered from what you capture, so photograph and note as you go.

FIELD KIT — BRING ON EVERY VISIT

Phone with GPS + camera — coordinates and photos in all directions

Tape measure / rangefinder — obstruction heights and standoff distances

Speed-test app — verify sustained up/down at the drop, not in the lobby

This guide — capture everything before you leave

WHAT YOU'RE DECIDING ON SITE

1 Can a dock live here? Clear takeoff/landing column, GPS sky view, a viable mounting surface, and safe access for install and service.	2 Can the site support it? Power and network at the dock and the DAA site — meeting spec, not just present.	3 What could stop us? Obstructions, airspace limits, structural or RF hazards. Any one can force a relocation or a No-Go.
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DESK ASSESSMENT (OUTSIDE-IN)

Run the Remote Operations Assessment from open sources first. Walk on site knowing the likely constraints — don't discover them there.

Why it matters: a desk assessment costs an hour; a wasted site visit costs a day. Airspace and connectivity problems are visible from your desk — if the address sits under an altitude cap or has no viable ISP, you want that conversation with the customer before anyone drives out.

01 Airspace classification

Class of airspace, controlled/uncontrolled, proximity to airports, existing waiver coverage FAA Facility Map grid.

02 Weather & flyability

Prevailing winds and seasonal patterns against platform flight limits.

03 Connectivity available

Fiber/ISP options at the address; Starlink viability if wired service is thin.

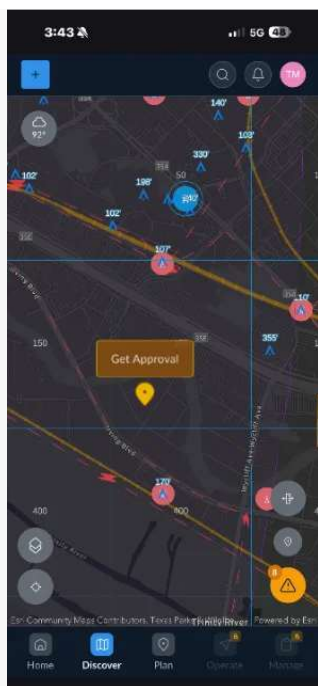
04 Site layout

Satellite and parcel view — building footprints, open areas, obstruction candidates.

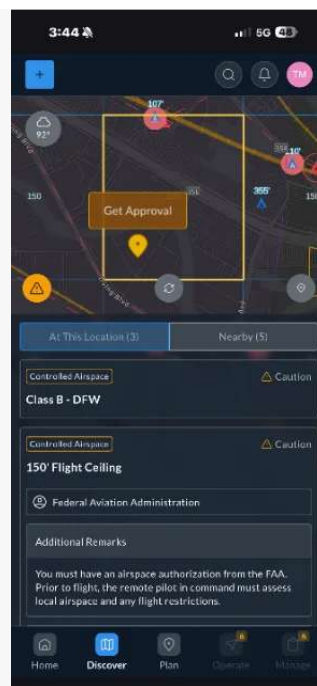


AIRHUB PORTAL
FIELD TOOL — PHONE APP

Search the site address and AirHub Portal returns the FAA Facility Map grid, the airspace classification, and nearby obstructions — run it during the desk assessment and again on site to confirm.



ADDRESS SEARCH — GRID & AIRSPACE CLASS



NEARBY OBSTRUCTIONS

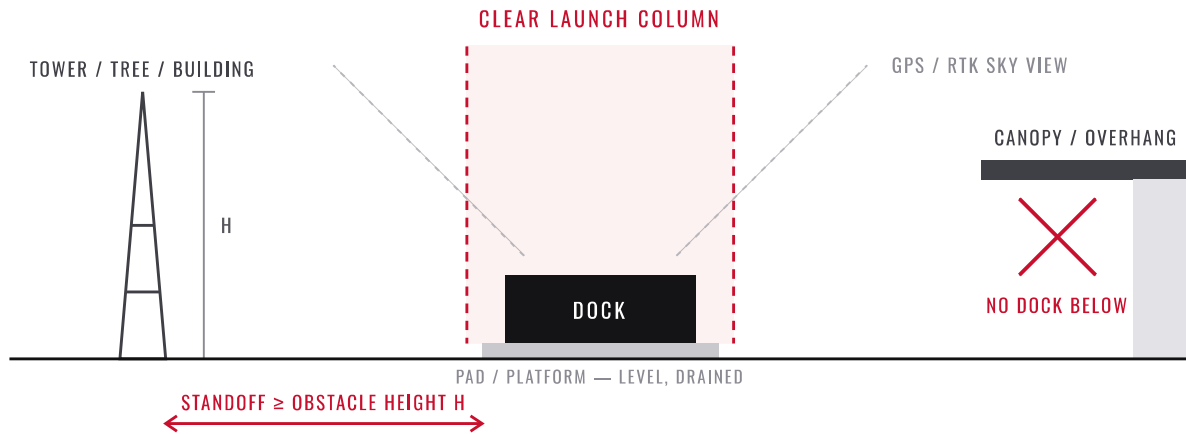


FIG. 1 — DOCK SITING CLEARANCES (SIDE VIEW)

- **Clear sky view.** Open overhead for GPS/RTK lock — no canopy or large overhang above the dock.
- **Standoff from tall obstacles.** Distance from towers, buildings, and large trees relative to their height.
- **Power and network within reach.** Dock sits near, or can be trenched/run to, the power and data drops.
- **Unobstructed launch column.** Vertical clear space for takeoff and landing, plus the immediate departure path.
- **Access for install & service.** Vehicle/equipment access to the location for setup and ongoing maintenance (roof, etc.).
- **Secure, controlled location.** Fenced or access-controlled area; not exposed to public foot traffic.

STEP 3 — ON SITE

MOUNTING & STRUCTURAL

The dock is an outdoor appliance that lives on site for years — it needs a surface that is level, stable, drained, and serviceable. Ground level is the default; rooftops buy sky view and security but add structural, grounding, and access questions that must be answered by the building owner's engineer, not by us on a walkthrough.

GROUND-LEVEL DOCK

- Flat level ground. Concrete, asphalt, leveled rock/road base.
- Concrete pad or pavers sized to the dock footprint
- Level, stable, and drains away from the unit

ROOFTOP DOCK

- Non-penetrating ballasted mount preferred on membrane roofs, with a rubber mat between the roof and the mount
- Service access to the roof for install and maintenance — stairs vs. ladder determines how docks get to the roof (crane, man lift, etc.); someone onsite checks the dock weekly, so access must be there

STEP 4 — SITE PROVIDES

POWER

Power is the customer's side of the deal, and it is the item most often left vague until install day. Your job on site is to find the nearest working outlet or panel, measure the distance to the dock location, and get the customer to commit — in writing, with their electrician — to one of the two options below if the run doesn't exist yet.

110V

STANDARD BUILDING POWER

A working 110V outlet at the dock site. Standard building power — no dedicated service required. If there's no outlet at the dock, the customer's **licensed electrician** provides one of the two options below.

OPTION A — TRENCH + OUTLET

Trench power out to the dock site and install a 110V outlet at the location.

OPTION B — JUNCTION BOX + WHIP

Terminate the circuit in a junction box, then trench the cable to the dock site per local code. Leave a 3-ft whip (wires stripped and capped) for DXD to land into our own connector.

NETWORK POWER

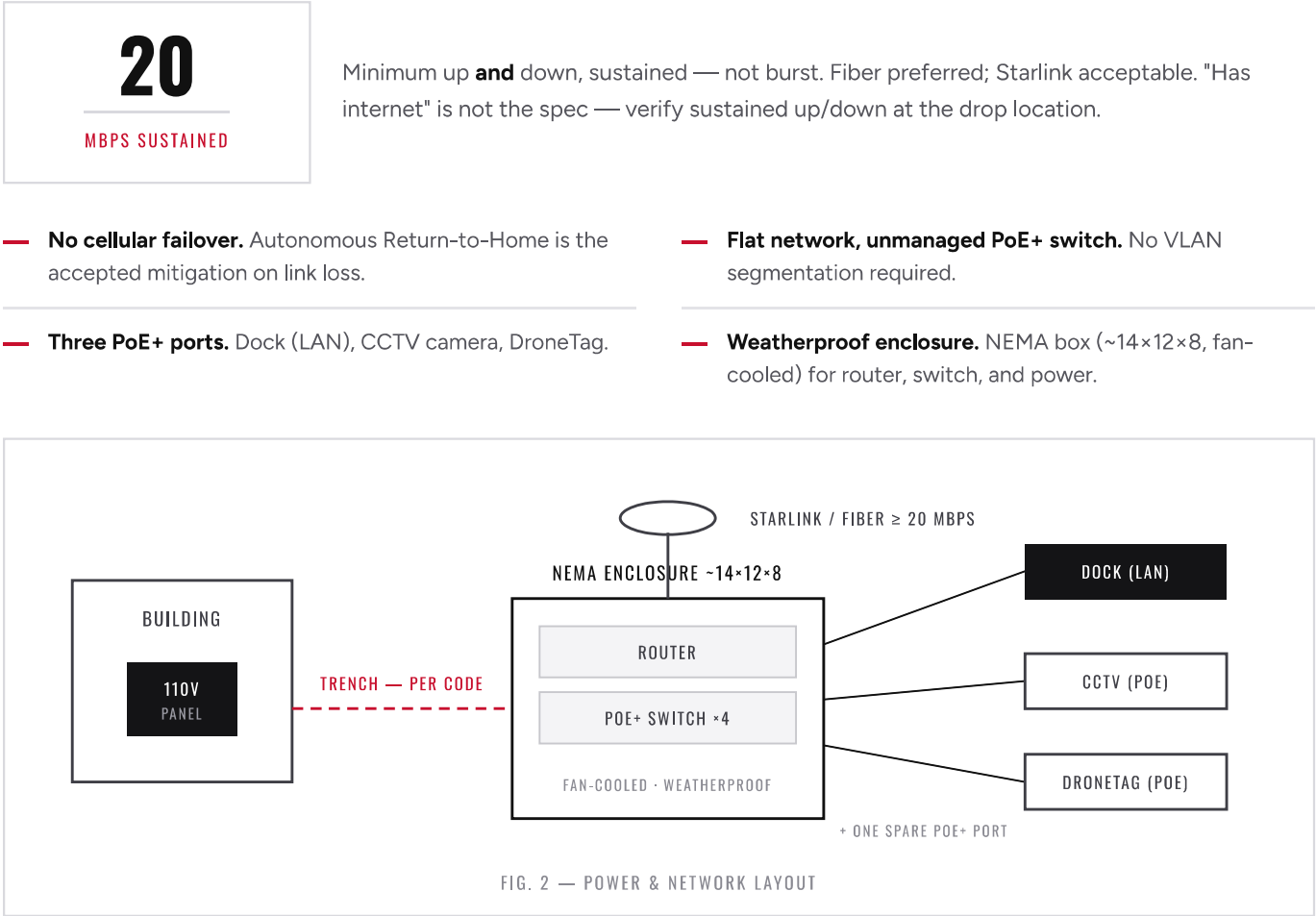
If DXD is providing the network, the customer also supplies power for that equipment.

DAA + CCTV

No separate feed — both run on PoE from the Starlink network kit. If customer is providing network, follow the instructions on the next step.

NETWORK & CONNECTIVITY

The link carries command-and-control and live video — it is the deployment's lifeline. Run a speed test at the actual drop location, at the time of day the system will operate, and record the numbers in the survey. A site that "has WiFi" and a site that sustains 20 Mbps both ways are different sites.



WHAT TO WATCH FOR

Any one of these can force a relocation or a No-Go. Photograph and note each one you find.

Why it matters: a red flag found on the walkthrough is a design input. The same flag found after hardware ships is a schedule slip, a re-survey, and a harder conversation with the customer. Finding problems is the job — write them all down, even the ones you think can be engineered around.

Tall obstructions

Towers, skyscrapers, and mature trees near the launch column or flight paths.

Overhead obstructions

Wires, canopies, and structures directly above the dock or departure path.

Airspace & waiver limits

Controlled airspace, altitude caps, or coverage the current waiver doesn't extend to.

RF interference

Nearby transmitters or dense RF that can degrade C2 link and GPS. RTK installation can be done, but will require a relocation of the DroneTag.

Structural / environmental

Weak mounting surfaces, poor drainage, corrosive or debris-heavy settings.

CONFIRM BEFORE INSTALL

CUSTOMER RESPONSIBILITIES

These are on the customer. Confirm each is in place or scheduled before an install date is committed.

ITEM	REQUIREMENT
Power to dock + DAA/RID site	110V
Network to dock + DAA/RID site	Ethernet, min 20 Mbps sustained up/down; DXD can provide Starlink at additional charge if customer cannot supply network.
Dock securing method	Level surface for the dock (concrete, asphalt, flat rooftop, leveled rock/road base, etc.)
Pole / tower for DAA / RID	Casia G requires a tower — DXD can coordinate an install at customer expense if none exists. DroneTag goes on a small pole near the dock; DXD will provide one if one does not exist.
Secure storage until install	Safe, controlled space to hold hardware between delivery and install. If not, equipment can be shipped to Dallas at our facility.

COMMON MISTAKES

Accepting "we have internet" as the network answer Burst speed, lobby WiFi, and shared guest networks all fail the spec. Hardwired ethernet speeds must be greater than 20 Mbps up and down.	Forgetting the DAA/RID site If power and network are required at two locations due to the RTK relay, make sure that is discussed.
Assessing only one candidate location Note at least a second viable spot while you're there. If engineering flags the primary, a backup already assessed saves a return trip.	

REFERENCE

GLOSSARY

DOCK	Weatherproof base station the drone launches from, lands on, and charges in	DAA	Detect-and-Avoid — sensors (e.g. Casia G) that watch for other aircraft near the operation
RID	Remote ID — broadcast of the drone's identity and position (e.g. DroneTag), required by regulation	GPS / RTK	Satellite positioning; RTK is the correction layer for centimeter-level precision landing — needs open sky
RTK RELAY	Ground station rebroadcasting RTK corrections across large urban sites; relocates the DroneTag to a second powered, networked location	C2 LINK	Command-and-control link between the drone and the operations center
RTH	Return-to-Home — autonomous fly-back and landing on link loss; the accepted failover mitigation	POE+	Power over Ethernet — one cable carries both power and data to CCTV, DroneTag, and dock LAN
NEMA BOX	Weatherproof-rated enclosure (~14×12×8, fan-cooled) housing router, switch, and power	WAIVER	FAA authorization the operation flies under; it defines coverage limits and altitude caps
WHIP	Short length of free cable (3 ft, stripped and capped) left at the dock for DXD to land into our connector		

GO / NO-GO

QUICK GATE BEFORE YOU LEAVE

01 Clear launch column overhead

02 Airspace workable under the waiver

03 Viable mounting surface

04 Power in reach (110V)

All four clear → Go. Any hard fail → relocate the dock or No-Go the site.

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